

Differential Equation

2. First Order First Degree

(a) Variable separable

Example: Verification of Solution: A very important aspect of solution of ODE is it must satisfy the ODE. This is so simple and obvious and yet so important.

Verify that $y = \frac{c}{x}$ (c an arbitrary constant) is a solution of the ODE $xy' = -y$

Solution: Differentiating, $y = \frac{c}{x}$ to get $y' = -\frac{c}{x^2}$; thus $xy' = -\frac{c}{x}$, $xy' = -y$, Hence ODE.

Example: Graphical Solution $y' = dy/dx = \cos x$

Solution: We will analyze the ODE using curves

The ODE $y' = dy/dx = \cos x$, $\int dy = \int \cos x dx$ (Using variable separable)

$y = \int \cos x dx = \sin x + c$, where c is an arbitrary constant.

This is a family of solutions. For $c = -3, -2, -1, 0, 1, 2, 3, 4$.

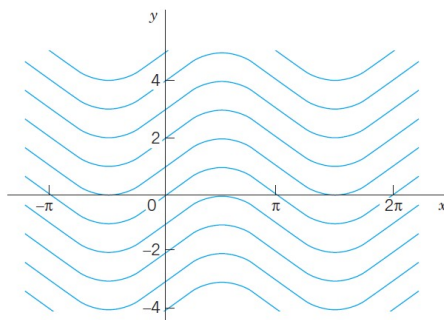


Fig. Family of solutions $y = \sin x + c$

Example: (A) Exponential Growth $y' = \frac{dy}{dt} = 0.2y$

Solution: The solution is $y = ce^{0.2t}$ can be calculated using variable separable.

This ODE is of the form $y' = ky$. With positive-constant k it can model exponential growth, for example colonies of bacteria or populations of animals. It also applies to humans for small populations in a large country (e.g., the United States in early times) and is then known as Malthus's law.

(B) Exponential Decay $y' = \frac{dy}{dt} = -0.2y$, Similarly, $y' = -0.2$ (with a minus on the right) has the

solution $y = ce^{-0.2t}$ for example a radioactive substance.

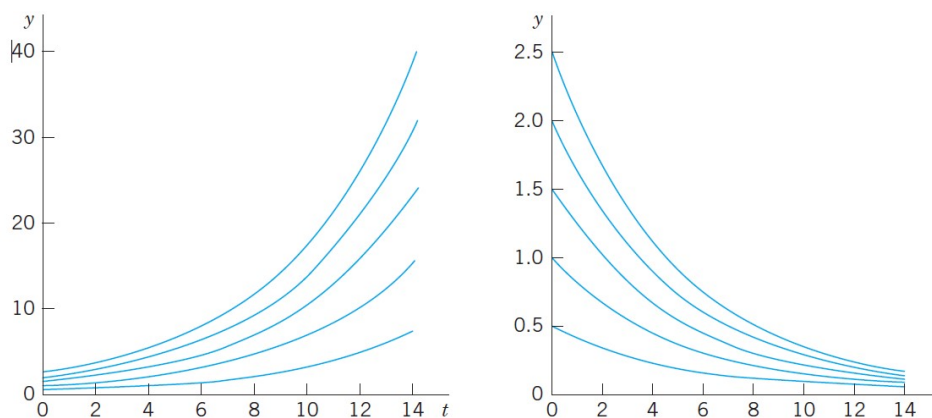


Fig. Soln. of $y' = 0.2y$ (exponential growth) Soln. of $y' = -0.2y$ (exponential decay)

Example: Initial value Problem

Solve the initial value problem

$$y' = \frac{dy}{dx} = 3y, \quad y(0) = 5.7$$

Solution: The general solution is $y(x) = ce^{3x}$ using variable separable. Apply initial condition; we obtain $y(0) = ce^0 = c = 5.7$. Hence, the initial value problem has the solution $y(x) = 5.7e^{3x}$.

This is known as particular solution.

Example: Radioactivity. Exponential Decay

$$\frac{dN}{dt} = -kN$$

where the constant k is positive, so that, because of the minus, we do get decay.

Solution: Using variable separable $N(t) = N(0)e^{-kt}$

Interpretation of result. Gives the amount of radioactive substance at time t . It starts from the correct initial amount and decreases with time because k is positive.

The limit of y as $t \rightarrow \infty$ is zero.

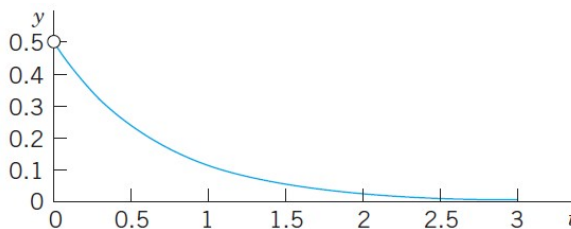


Fig. Radioactivity (Exponential decay)

Singular solution: An ODE may sometimes have an additional solution that cannot be obtained from the general solution and is then called a singular solution.

Example: The ODE $y'^2 - xy' + y = 0$ is of this kind.

Check $y = cx - c^2$ and the singular solution $y = x^2 / 4$

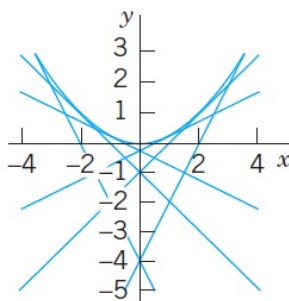


Fig: Particular solutions and singular solutions

Example: Solve the differential equation

$$9yy' + 4x = 0$$

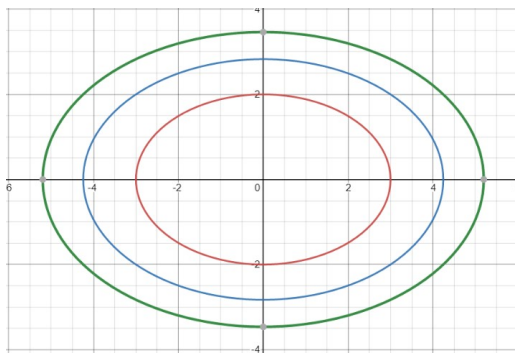
Solution: By separating variables, we have

$$9ydy = -4x dx$$

By integrating on both sides, we obtain the general solution

$$\frac{9}{2}y^2 = -2x^2 + c^2, \quad \frac{x^2}{9} + \frac{y^2}{4} = c$$

The solution represents a family of ellipses.



Example: Solve the differential equation

$$y' = 1 + y^2.$$

Solution: By separating variables and integrating we obtain

$$\frac{dy}{1+y^2} = dx, \tan^{-1} y = x + c, y = \tan(x + c)$$

Example: Exponential growth or decay

Exponential growth or decay is governed by the differential equation

$$y' = ky$$

with positive or negative (fixed) k , respectively.

Solution: Equation is Variable separable.

$$\frac{dy}{y} = k dx, \quad \ln|y| = kx + c$$

because by calculus, $(\ln|y|)' = y'/y$; when $y > 0$, then $(\ln|y|)' = (\ln y)' = y'/y$; and when $y < 0$, then $-y > 0$ and $(\ln|y|)' = -y'/(-y) = y'/y$. Taking exponentials and noting that $e^{a+b} = e^a e^b$, we get

$$|y| = e^{kx+c} = e^{kx} e^c, \quad \text{Thus, } y = ce^{kx}$$

Where new constant $c \Rightarrow +e^c$ when $y > 0$; and $c \Rightarrow -e^c$ when $y < 0$, and we can also admit $c = 0$. (Giving $y = 0$). This is the general solution.

Example: Initial value problem

Solve the initial value problem

$$y' = -\frac{y}{x}, \quad y(1) = 1.$$

Solution: Separation of variables and integration gives

$$\frac{dy}{y} = -\frac{dx}{x}, \quad \ln|y| = -\ln|x| + \tilde{c} = \ln\frac{1}{|x|} + \tilde{c}.$$

Taking exponentials, we get $y = c/x$. This is the general solution. From this and the initial condition, $1 = c/1$, so that $c = 1$. The answer is $y = 1/x$.

Example: Initial value problem. Bell-shaped curves

Solve the initial value problem

$$y' = -2xy, \quad y(0) = 1.$$

Solution: Separating variables, integrating, and taking exponentials, we obtain

$$\frac{dy}{y} = -2x dx, \quad \ln|y| = -x^2 + \tilde{c}, \quad |y| = e^{-x^2+\tilde{c}}.$$

Setting $e^{\tilde{c}} = +c$ when $y > 0$, and $e^{\tilde{c}} = -c$ when $y < 0$, and admitting also $c = 0$ (which gives the solution $y = 0$), we get the general solution

$$y = c e^{-x^2}$$

A Bell-Shaped Gaussian curve $y(0) = 1$.

$$y = e^{-x^2}$$

